

Sensor Data analysis Report

Chart Title: Raw vs. Cleaned Sensor Data Trend Over Time

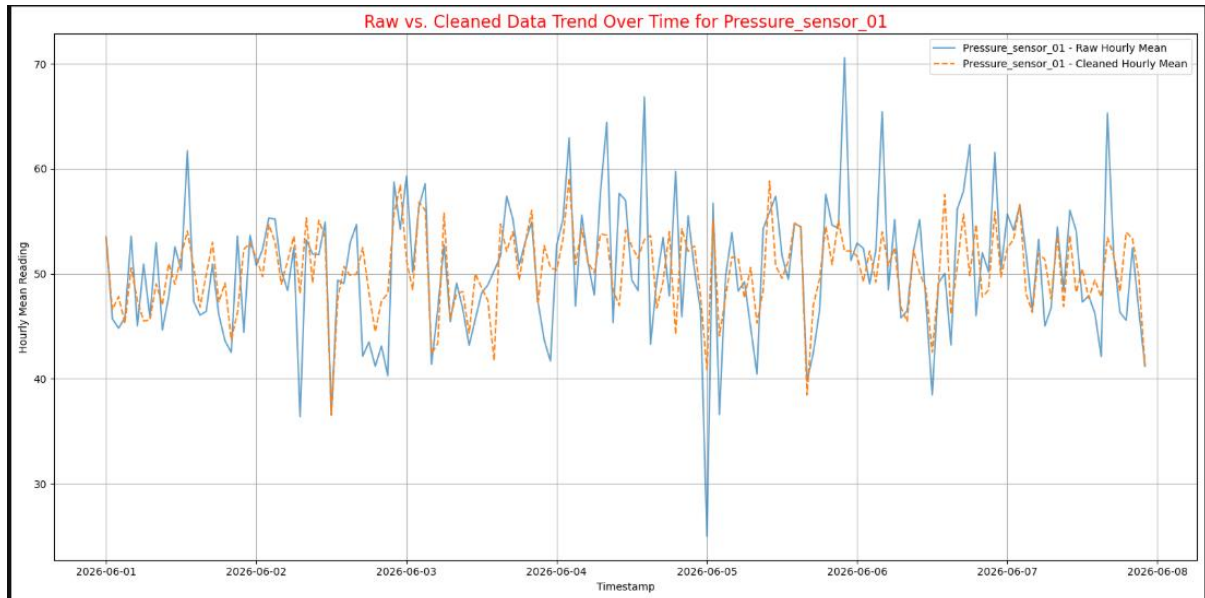


Chart Annotations

Raw Data Spike: "Outlier created a false pressure spike, leading to misleading maintenance alerts."

Missing Data: "Missing readings hid the true operating trend."

Cleaned Data: "Cleaning revealed a gradual increase in pressure over time."

Context

The original sensor data contained extreme outlier values that distorted the actual performance of the system. These data quality issues created false pressure spikes and irregular fluctuations, making it difficult to distinguish genuine equipment behaviour from measurement errors. Relying on the raw data could result in unnecessary maintenance actions or cause operators to overlook developing operational issues.

Insight

After cleaning the data, the true pressure trend became clear. Instead of appearing stable, the system showed a gradual pressure increase during later operating hours, revealing a potential operational concern.

Action

Immediate action should be taken to strengthen monitoring during periods when pressure begins to rise. Maintenance inspections should be scheduled before the identified high-pressure period, and automated alerts should be configured using the cleaned data rather than the raw sensor readings. This will improve the accuracy of operational decisions, reduce false maintenance alarms, minimize unplanned equipment downtime, and support safer, more reliable plant operations.